

## Education

- 2021–2025 : **Ph.D. in Computer Engineering**, *Northeastern University*, Boston, USA.  
2018–2020 : **Masters of Science in Computer Engineering**, *Southern Illinois University Carbondale*, IL.

## Skills

- Languages Python, C++, Java/Android, Bash, SQL, HTML, CSS
- Tools XCAL-Accuver (5G/LTE signaling message analysing tool) || Keysight RuSIM || Spectrum Analyzer || Git || Wireshark || Scikit-Learn || PyTorch || Keras || SoA Machine-Learning Frameworks || Video Streaming tools (Puffer, FFmpeg)
- Platform Linux || Android || Docker || Cloud (AWS, GCP) || O-RAN
- Protocols NR/LTE 3GPP standards || TCP/MPTCP/UDP protocols and their implementation (Linux Source Codes) ||
- /Standards IEEE 802.11 ax/ac/ad/b/g/n standards

## Experience

- 2026 – 2026 **Research Engineer, AT&T Labs. Inc.**, contract.
- Developed a production-ready anomaly detection pipeline for E911 network KPIs using a lightweight Isolation Forest model, enriched with SIP/CCDR protocol context, including message types, cause codes, and failure indicators. Engineered historical baseline features using z-scores, rolling means, and rolling standard deviations to identify KPI degradation, achieving 90% anomaly correlation with known network incidents.
  - Benchmarked graph-based SSB optimization approaches using Google OR-Tools and SCIP on production network inventory data, incorporating neighbor weights to evaluate optimization performance in terms of execution time, number of configuration changes, and iterations required to reach optimality.
- 2021 – 2023 **Research Intern and External Collaborator, AT&T Labs. Inc.**
- Investigated the impact on QoE of low-latency video streaming over 5G networks in a joint collaboration work with Facebook.
  - Published a peer-reviewed conference paper correlating radio link dynamics with application-layer KPIs to proactively mitigate User Quality of Experience (QoE) degradation.

## Selected Projects

### **X5G: O-RAN compliant 5G Testbed Development, NSF Funded.**

- Contributed to the development of X5G: an O-RAN compliant multi-vendor (NVIDIA, OAI, Foxconn) programmable testbed for research.
  - Instrumented OAI code for performance tuning, configuration tweaking, and KPI extraction.
  - Developed measurement report and event (A3) handling mechanism for Xn-handover.
  - Improved single UE user throughput by 16% by debugging and bug reporting to Nvidia Aerial and OAI team.
  - Published 1 Conference and 1 Journal Paper.
- OAI  
C++  
O-RAN  
NVIDIA ARC

### **5G measurement studies on user-perceived performance, NSF Funded.**

- Designed and executed large-scale mobility measurement campaigns (multi-city/cross-country drives), integrating synchronized application-level tests with cellular KPIs for holistic analysis.
  - Assessed 5G coverage, throughput, latency, and user Quality of Experience (QoE) using large-scale dataset [10TB+].
  - Built an end-to-end measurement and QoE extraction pipeline for modern interactive applications, including AR/VR, 360° video streaming, and cloud gaming.
  - Developed tools to extract lower-layer (PHY/MAC/RLC) KPIs from 5G control-plane signaling (RRC) [XCAL logs].
  - Investigated key 5G mechanisms—handovers, uplink power control, and advanced features (carrier aggregation, beamforming, MIMO) to identify performance bottlenecks and explain observed behavior.
  - Built automated Python pipelines for data cleaning, feature extraction, and statistical analysis; produced publication-quality figures and reproducible artifacts.
  - Open-sourced **datasets/scripts** and created replication-friendly workflows to enable follow-on studies.
  - Published 10 conference papers.
- 5G/LTE  
Python  
X-CAL  
Puffer  
FFMPEG

### **Throughput prediction in operational 5G networks, CISCO Funded.**

- 5G/LTE
  - Collected a fine-grained, real-world **dataset** during a 4-day (1000+ km) drive, measuring bidirectional application throughput at 100 ms time-scales across all three major U.S. operators and mixed LTE/5G deployments.
- Python
  - Evaluated multiple predictors (MLP, GDBT, transformer-based MVTs) alongside statistical baselines (EWMA, last-value) and showed that low RMSE can still yield poor application-centric accuracy (high ARE95 / low PARE10).
- Machine Learning
  - Used over-parameterized neural networks as a “learnability test” and found slow training convergence and persistently high validation/test error, indicating fundamental limits to fine-grained throughput predictability.
  - Identified conflicting sample sequences, where nearly identical historical throughput patterns were followed by drastically different next-sample values, forcing models to learn highly irregular mappings and limiting their generalization performance.
  - Published 1 Conference paper.

### **5G Performance Evolution Study, NSF Funded.**

- JAVA
  - Built an IRB-approved **Android measurement app** (NextG-UP) to collect uplink TCP throughput and RTT alongside cellular KPIs (GPS, RSRP/RSRQ/RSSI, Cell-ID, ARFCN/EARFCN, operator); implemented nuttcp (10s UL test) + ICMP ping (11 packets @200ms) with a lightweight footprint (6.5MB APK, <250MB RAM).
- Android
  - Deployed a geographically distributed measurement backend (AWS servers in N. Virginia, Oregon, Frankfurt + AWS Wavelength inside Verizon’s network in Boston) and integrated nearest-server selection + mobility modes (static/walking/driving).
- Python
  - Led a cross-sectional, year-long 5G uplink campaign with weekly volunteer measurements: 10,166 tests across 8 cities in 7 countries (2 continents), covering 12 operators and 1,422 unique cell IDs.
- AWS
  - Published 2 Conference paper and 1 Journal Paper.
- Machine Learning

### **Multipath TCP (MPTCP) for Dual-band 802.11ad + 802.11 ac networks, NSF Funded.**

- C
  - Ported MPTCP source code to a commercial dual-band smartphone (802.11ad@60 GHz + 802.11ac@5 GHz) by integrating 15,000+ LOC into the Android Linux kernel; released the implementation publicly for reproducible experimentation.
- Python
  - Built an experimental pipeline to measure **100 ms**-granularity throughput and on-device power draw, enabling a first look at both *performance* and *energy* of MPTCP on resource-constrained phones.
- TCP
  - Identified and implemented key system-level optimizations for Gbps MPTCP on smartphones (e.g., enabling GRO and CPU/SIRQ affinity to big cores), achieving near sum-rate performance across the two radios under ideal conditions.
- MPTCP
  - Quantified throughput–energy tradeoffs across packet-scheduling ratios and derived practical guidelines toward an energy-aware MPTCP scheduler for dual-band mmWave/legacy Wi-Fi phones.
- Android
  - Released two custom kernels with MPTCP stack for **ASUS ROG II** and **Pixel 5**.
- Kernel
- tc
- 802.11

## **Selected Publications**

### **In Conference Proceedings**

- 2024 On the Predictability of Fine-grained Cellular Network Throughput using Machine Learning Models. In *Proc. of IEEE MASS*, 2024.
- 2024 An Experimental Study of Low-Latency Video Streaming over 5G. In *Proc. of IEEE MeditCom*, 2024.
- 2023 Performance of Cellular Networks on the Wheels. In *Proc. of ACM IMC*, 2023.
- 2021 Multipath TCP in Smartphones Equipped with Millimeter Wave Radios. *Proc. of ACM WiNTECH '21*, 2021.

### **Journal Articles**

- 2025 X5G: An Open, Programmable, Multi-Vendor, End-to-End, Private 5G O-RAN Testbed With NVIDIA ARC and OpenAirInterface. *IEEE Transactions on Mobile Computing*, 2025.
- 2025 How mature is 5G deployment? A cross-sectional, year-long study of 5G uplink performance. *Computer Communications*, 2025.

## **Professional Services**

- 2026 COMCOM, Computer Networks, IEEE Transactions on Green Comm. and Networking, *Reviewer*.
- 2025 COMCOM, IEEE VTC, IEEE Transactions on Green Comm. and Networking, *Reviewer*.
- 2024 Computer Networks, IEEE IoT, IEEE ICC, IEEE WCNC, *Reviewer*.
- 2024 IEEE LANMAN, IEEE WOWMOM, *Web Chair*.
- 2023 IEEE WCNC, *TPC Member*.